

Modeling Healthcare Capacity: Extending the SIR Framework

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December 2025

1. Introduction

Mathematical models are essential tools for predicting and understanding the spread of infectious diseases. Among these, the Susceptible-Infectious-Recovered (SIR) framework has served as a primitive approach to simulating epidemic growth and decline. The standard SIR model divides a population into three categories: susceptible individuals who can become infected, infectious individuals who can infect others, and recovered (or removed) individuals who gain immunity. Its dynamics are driven by two parameters: the transmission rate β and the recovery rate γ .

The standard SIR model assumes that all infected individuals recover at a constant rate, immunity is perfect after infection, all individuals are equally susceptible, and all individuals experience an equally severe infection. These simplifying assumptions make the model mathematically manageable, but they neglect crucial real-world dynamics.

In real epidemics like COVID-19, access to medical care determines whether infected individuals recover or die. The availability of hospital resources - including ward beds, ICU beds, ventilators, treatments, and healthcare workers - determine the outcome for each individual. The classic SIR model cannot account for this, as it lumps deaths into the “removed” category and does not differentiate between mild and severe illness. As a result, this model cannot address vital healthcare considerations such as: How does hospital overload influence recovery rates, mortality, and outbreak cycles? How does treatment prioritization or ICU scarcity alter outcomes across age groups? How do waiting times influence severe outcomes and deaths?

Previous studies have highlighted the limitations of excluding hospital considerations from infectious disease models. For example, projections of hospital demand from early COVID-19 showed ICU capacity could be exceeded within weeks, with mortality climbing as care availability declined [1]. Another analysis showed that the recovery rate γ falls as hospitals fill, directly contradicting the constant γ assumption of the classic SIR model. Furthermore, the hospital bed parameter also caused complex epidemic trajectories - including sudden shifts between controlled and uncontrolled outbreaks, and scenarios where an epidemic persists even when the basic reproduction number R_0 is below one [2]. These findings

illustrate that (1) hospital overload is not hypothetical and (2) R_0 alone does not determine epidemic outcomes; capacity constraints play a pivotal role.

To more accurately model healthcare limitations and severe disease, epidemiological models must move beyond the traditional SIR framework. Modeling hospitalization requires new compartments that represent: (1) patients requiring care, (2) patients admitted to the hospital, and (3) patients who are forced to wait for admission due to oversaturation. The new model must also distinguish between ICU and ward care, account for differences in disease severity and susceptibility across age groups and occupations, incorporate vaccination and treatment strategies, and separately track deaths and recoveries.

This extended approach allows us to better understand scenarios that reflect real healthcare applications where capacity restrictions, staffing shortages, and treatment options shape epidemic outcomes. Incorporating healthcare considerations into infectious disease models provides a more realistic framework for policymakers to anticipate epidemic trajectories so that they can design the appropriate interventions to reduce mortality rates. Through this lens, we aim to identify practical strategies to minimize deaths and prevent hospital oversaturation during future epidemics.

2. Methods

To investigate the impact of hospital capacity on epidemic dynamics and mortality, we developed a deterministic, age-structured compartmental model extending the classical SEIR framework. Our model, designated SEIXHRD (Susceptible-Exposed-Infected-Severe-Hospitalized-Recovered-Deceased), explicitly incorporates healthcare system constraints by differentiating between patients receiving care and those awaiting admission due to capacity saturation.

2.1. Model Structure and Compartments

The population is divided into N age groups to account for age-dependent contact rates and disease severity. Each of these age groups is further split into the following compartments:

- S : Susceptible individuals who are not yet infected.

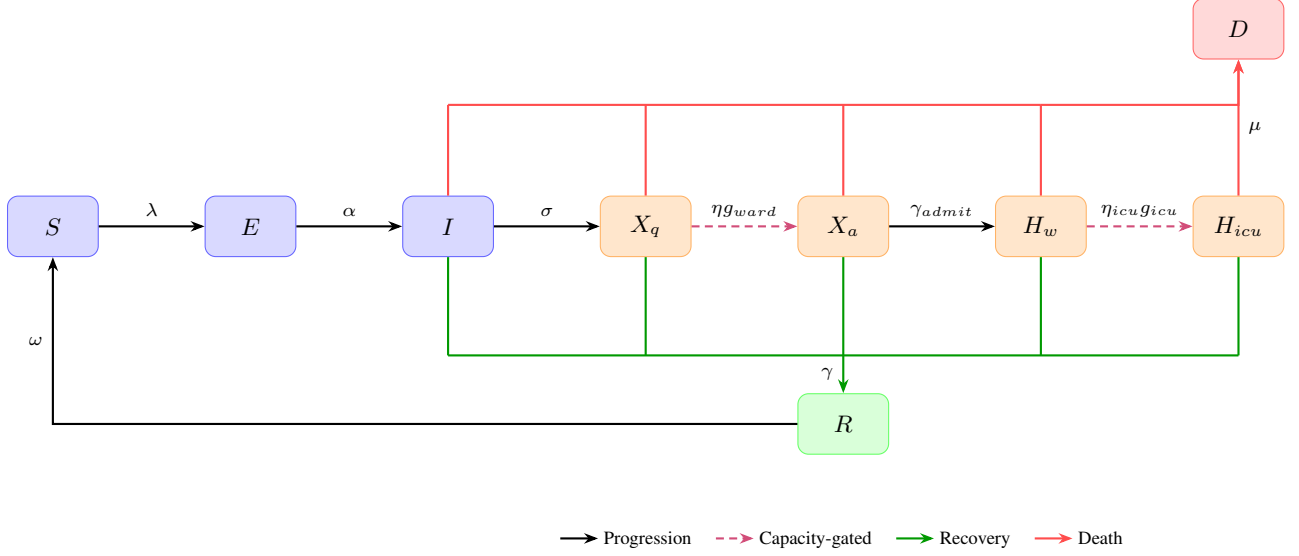


Figure 1: Compartmental flow diagram of the SEIXHRD model. S : Susceptible, E : Exposed, I : Infected, X_q : Severe (queued), X_a : Severe (admitted), H_w : Hospitalized (Ward), H_{icu} : Hospitalized (ICU), R : Recovered, D : Dead. Purple dashed arrows indicate capacity-constrained transitions governed by Hill function gating (g_{ward} , g_{icu}).

- E : Exposed individuals who have been infected but aren't yet infectious.
- I : Infectious individuals
- X_{queued} : Queued individuals with severe disease who need hospitalization and are waiting for a bed.
- $X_{admitted}$: Individuals with severe disease who have been admitted but are not yet receiving full care, with a slightly reduced mortality rate.
- H_{ward} : Individuals with a ward bed, which reduced mortality rates even more.
- H_{icu} : Individuals with an ICU bed receiving the most critical care and the lowest mortality rates.
- R : Recovered individuals who have survived and gained immunity (either temporary or permanent).
- D : Cumulative deaths that are tracked separately between those who received care $D_{treated}$ and those who did not $D_{untreated}$.

2.2. Model Flowchart

Figure ?? illustrates the compartmental structure and transition pathways of the SEIXHRD model. Arrows indicate flow through the system, green arrows indicate recovery flow, and red arrows denote mortality. The capacity-gated transitions (controlled by g_{ward} and g_{icu}) are highlighted by purple dashed arrows. This system assumes a closed population and doesn't account for the

birth rate and natural death rate. Vaccinated compartments and age-structuring are also left out to reduce noise. A flow for waning immunity from R back to S is included.

2.3. Capacity Constraints and Gating

A central feature of this model is the implementation of smooth capacity constraints rather than hard cutoff thresholds. In our model, the admission rates degrade non-linearly as hospital occupancy approaches its capacity.

To do this, a Hill function was utilized to model the probability of admission, $g(H)$, defined as:

$$g(H) = \frac{1}{1 + (\frac{H}{K})^n}$$

Where H is the current occupancy, K is the hospital capacity (combined ward and ICU bed counts), and n is the Hill coefficient that determines the steepness of the cutoff. This gating function controls the flow from X_{queued} to $X_{admitted}$ and from H_{ward} to H_{icu} . When occupancy is low ($H \ll K$), $g(H) \approx 1$, allowing normal admission flow. As H approaches K , $g(H)$ drops rapidly, causing patients to accumulate in the X_{queued} compartment where they face higher mortality rates.

2.4. Varying Mortality

To capture the true impact of hospital collapse, we assigned different mortality rates based on care status. The mortality rate for the queued compartment ($\mu_{untreated}$) is significantly higher than that of the admitted compartment

($\mu_{treated}$), reflecting the lack of life-saving interventions like intravenous fluids and ventilators.

2.5. Differential Equations

The model dynamics are governed by a system of ordinary differential equations for each age group i . The system was solved using `scipy.integrate.solve_ivp`. To avoid listing every age group individually, the age index is implied in the following equations.

2.5.1 Force of Infection

The force of infection $\lambda(t)$ represents the per-capita rate at which susceptible individuals become infected:

$$\lambda_i(t) = \beta(t) \sum_j \frac{C_{ji} \cdot I_{effj}}{N_j}$$

where C_{ji} is the contact matrix element representing contacts made by age group j with age group i , and N_j is the population of age group j . The effective infectious population I_{eff} incorporates relative infectiousness modifiers:

$$I_{eff} = (I + \theta_X X_{total}) + \theta_{vax}(I_{vax} + \theta_X X_{vax_{total}}) + \theta_H H_{total}$$

where θ_X , θ_H , and θ_{vax} are infectiousness modifiers for severe cases, hospitalized patients, and vaccinated individuals, respectively.

2.5.2 Unvaccinated Compartment Dynamics

The flow through unvaccinated compartments is governed by:

$$\begin{aligned} \frac{dS}{dt} &= \nu N_{total} - \lambda S + \omega R - \phi S - \mu_{bg} S \\ \frac{dE}{dt} &= \lambda S - \alpha E - \mu_{bg} E \\ \frac{dI}{dt} &= \alpha E - (\gamma_I + \mu_I + \sigma) I - \mu_{bg} I \end{aligned}$$

Severe disease compartments incorporate capacity gating:

$$\begin{aligned} \frac{dX_{queued}}{dt} &= \sigma I - (\gamma_X + \mu_{X_{untreated}}) X_{queued} \\ &\quad - \eta \cdot g_{ward} \cdot X_{queued} - \mu_{bg} X_{queued} \\ \frac{dX_{admitted}}{dt} &= \eta \cdot g_{ward} \cdot X_{queued} - (\gamma_X + \mu_X) X_{admitted} \\ &\quad - \gamma_{admit} X_{admitted} - \mu_{bg} X_{admitted} \end{aligned}$$

Hospital dynamics with ICU gating:

$$\begin{aligned} \frac{dH_{ward}}{dt} &= \gamma_{admit} X_{admitted} - (\gamma_{ward} + \mu_{ward_{eff}}) H_{ward} \\ &\quad - \eta_{icu} \cdot g_{icu} \cdot H_{ward} - \mu_{bg} H_{ward} \\ \frac{dH_{icu}}{dt} &= \eta_{icu} \cdot g_{icu} \cdot H_{ward} - (\gamma_{icu} + \mu_{icu}) H_{icu} \\ &\quad - \mu_{bg} H_{icu} \end{aligned}$$

Recovery and death accumulation:

$$\begin{aligned} \frac{dR}{dt} &= \gamma_I I + \gamma_X (X_{queued} + X_{admitted}) \\ &\quad + \gamma_{ward} H_{ward} + \gamma_{icu} H_{icu} - \omega R - \mu_{bg} R \\ \frac{dD}{dt} &= \mu_I I + \mu_{X_{untreated}} X_{queued} + \mu_X X_{admitted} \\ &\quad + \mu_{ward_{eff}} H_{ward} + \mu_{icu} H_{icu} \end{aligned}$$

2.5.3 Effective Ward Mortality

The effective ward mortality rate $\mu_{ward_{eff}}$ accounts for excess deaths when ICU capacity is saturated:

$$\mu_{ward_{eff}} = \mu_{ward} + (\mu_{ward_{denied}} - \mu_{ward}) \cdot \eta_{icu} \cdot (1 - g_{icu})$$

This formulation ensures that ward patients who require but cannot access ICU care experience elevated mortality rates proportional to the degree of ICU saturation.

2.5.4 Vaccinated Population Dynamics

A parallel compartmental system exists for vaccinated individuals ($S_{vax}, E_{vax}, \dots$). The dynamics are identical in structure, but parameters are modified by vaccine efficacy factors:

- **Infection:** $\lambda_{vax} = \lambda \cdot (1 - VE_I)$
- **Severe disease:** $\sigma_{vax} = \sigma \cdot (1 - VE_S)$
- **Mortality:** $\mu_{vax} = \mu \cdot (1 - VE_D)$

2.5.5 Parameter Definitions

Key model parameters include:

- ν : Birth rate (inflow to S)
- α : Latent period rate ($E \rightarrow I$)
- σ : Severe disease progression rate ($I \rightarrow X$)
- η : Hospital admission rate ($X_{queued} \rightarrow X_{admitted}$)
- γ_{admit} : Ward transfer rate ($X_{admitted} \rightarrow H_{ward}$)
- η_{icu} : ICU need rate ($H_{ward} \rightarrow H_{icu}$)

- ϕ : Vaccination rate ($S \rightarrow S_{vax}$)
- ω : Natural immunity waning rate ($R \rightarrow S$)
- μ_{bg} : Background (non-disease) mortality rate
- g_{ward} : Hill function gating factor
- g_{icu} : Hill function gating factor

3. Results

4. Discussion

The primary goal of developing this model was to study and evaluate how hospital capacity limitations influence epidemic outcomes, specifically mortality and recovery rates. The simulations using the SEIXHRD model showed that fixed healthcare capacity creates a critical “tipping point” in epidemic dynamics that are not observable in standard SIR modeling.

4.1. Impact of Capacity Saturation

Our results demonstrate that hospital capacity acts as a non-linear threshold for mortality. In scenarios where the peak infection demand remained below capacity ($H < K$), the infection fatality rate (IFR) remained constant and relatively low. However, once the “surge” exceeded capacity, mortality rates did not increase linearly but rather spiked aggressively. This supports the hypothesis that the “flatten the curve” strategy is mechanistically valid not just for delaying infections, but for reducing the total area under the mortality curve. As shown in our comparison of hospital burden strategies, increasing capacity provides diminishing returns once the system is well-resourced, but for resource-limited settings (e.g., rural systems), small increases in capacity showed significant reductions in mortality.

4.2. The Hidden Mortality of Waiting Times

By separating the “Severe” compartment into X_{queued} and $X_{admitted}$, our model shows the hidden cost of wait times. In high-transmission scenarios ($R_0 > 2.5$), the bottleneck wasn’t always the total number of beds, but the rate of flow into them. The queue accumulation resulted in a significant number of “untreated” deaths.

4.3. Policy Implications and Interventions

Our analysis of intervention strategies highlights that early, strong interventions are far superior to delayed or cyclical strategies in preserving hospital function. The “Delayed Strong” strategy often resulted in a secondary peak that still overwhelmed ICU capacity because the initial seeding was

already too widespread. This suggests that interventions must be called by early indicators like transmission rates rather than late indicators like hospital occupancy overflow to be effective.

4.4. Model Limitations

While the SEIXHRD model attempts to add as much realism as possible, there are many limitations that prevent it from actually being useful for predicting real world dynamics. The model lacks the ability to effectively simulate healthcare worker capacity as well, something that definitely matters when attempting to model the impact hospitalization has on epidemics. Even if the hospitals have beds available, there needs to be healthy staff to operate the facilities. This, alongside parametrization with real-world epidemic data would give the model significantly more credibility.

5. Conclusion

We successfully demonstrated that incorporating hospitalization with capacity constraints into the SIR framework fundamentally alters the predicted infection fatality rate. The model confirms that mortality is not an intrinsic property of the virus alone but a dynamic variable dependent on the operational capacity of the healthcare system. Consequently, public health models that neglect capacity constraints likely underestimate death tolls in surge scenarios.

GitHub Repository

All the code for the model is available on GitHub:

<https://github.com/JasonHunter95/hospital-model>

References

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